Implementation of the ACTAM TPC in Geant4

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This document outlines the simulation process of the ACTAM TPC. It discusses the relevant files and classes, the definition of electron drift, and the simulated output.

For a detailed description of the simulation or event reconstruction procedure, including experimental motivation, geometry, material specifications, and preliminary results from the simulation, see the relevant chapters of my thesis.

1 The ACTAM TPC

The simulation supports the use of either helium-3 or helium-4 as an active gas, with pressure varying from 5 to 30 bar. Both of these parameters are set in the file *data/TPC.dat*, along with the dimensions of geometric components. The default version of this file is shown below.

```
## Anode specifications in mm
## Thickness : 1.5
## Distance of anode from cell end : 25
## Angular sections : 16
## Radius of anode central pad
## Radius of anode ring
## Radius of first , second , and third rows
## Radius of fourth row = outer radius of anode
Anode-Dim: 1.5 25 16 5 10 20 30 40 50
##
## Cathode specifications in mm
## Thickness : 1.0
## Radius : 50
## Distance to cathode from cell end : 25
Cathode-Dim: 1.0 50 25
##
##Grid specifications in mm
## Wire diameter : 0.05
## Space between wires : 0.5
Grid-Dim: 0.05 0.5
##
##Helium specifications
##Isotope: He3 or He4
##Pressure: approximated for anywhere from 20 to 30 bar
Helium: 3 25
##
## Run Mode 0=no, 1=yes
## Check overlaps
```

Run–Mode: 0 1

2 Electron Drift

To generate functional results, custom definition of electron drift was required. The constants used were simulated in **Magboltz**, and are implemented in the function *A2DriftModel::SetConstants*.

The electron drift model records the initial position of each produced electron, then sets its z coordinate to be in line with the anode. Final (x,y) coordinates as well as travel time are picked by a random engine from a Gaussian distribution. If the (x,y) positions are such that the electron hits the anode, a hit is recorded in the sensitive detector. The Gaussians used are based on work by Fabian Metzger [1]:

Position:

$$\mu_x = x_0, \mu_y = y_0$$

$$\sigma_x = \sigma_y = D_T\sqrt{|z_0 - z_{anode}|}$$

- Mean is initial (x,y) position of simulated electron
- Standard deviation depends on transverse diffusion coefficient and distance to anode

Time:

$$\mu_t = \frac{|z_0 - z_{anode}|}{v_{drift}}$$

$$\sigma_t = \frac{D_L}{v_{drift}}\sqrt{|z_0 - z_{anode}|}$$

- Mean is distance to anode divided by drift velocity
- Standard deviation depends on lateral diffusion coefficient, drift velocity, and distance to anode

3 Event Reconstruction

Four branches have been added to the **A2Geant4** output tree to hold the TPC data (Table 1). These branches are then used in event reconstruction, which must then be compared to the true event parameters. Tables 2 and 3 show how event data is stored in Geant4 and how the relevant parameters can be extracted.

To do event reconstruction, the analysis script *Reconstruct.C* can be used on any **A2Geant4** output file. The drift velocity, anode structure, and calibration values for the charge-energy relationship must be set manually in this file.

The charge-energy relationship is relatively constant at different pressures and isotopes of helium, but if any changes are made to the drift model this will have to be run again. The macro *TPCcalibrate.mac* generates calibration files, which can then be used with the analysis script *Calibrate.C* to find the new constants. These constants must then be changed in *Reconstruct.C*, or the *Analyze* class of **a2GoAT**.

Branch	Type	Quantity	Elements	Units
ntpc	integer	Number of hits in TPC anode	-	-
itpc	integer array	ID of anode section hit	Per hit	-
qtpc	float array	Charge deposited in hit	Per hit	e
ttpc	float array	Time of hit	Per hit	ms

Table 1: Branches added to the **A2Geant4** output tree to store TPC readout. The ID numbers stored in the branch *itpc* are those presented in Figure 1.

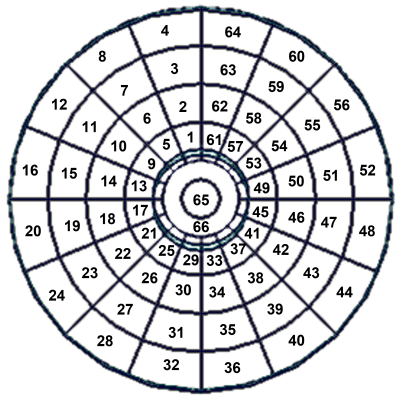

Figure 1: Copy numbers of the TPC anode. Note that the numbering is constant even as the dimensions are changed between anode designs.

Branch	Type	Quantity	Units
klab	float array	Kinetic energy of tracked particles	MeV
vertex	float array	Event position coordinates	cm
dircos	float array	Direction of motion of tracked particles	-

Table 2: **A2Geant4** branches storing true kinematic parameters - useful in event reconstruction.

Parameter	True value
Recoil energy	klab[0]
Event z position	vertex[2]
Recoil theta	acos(dircos[0][2])

Table 3: Recovering true event information from the **A2Geant4** output.

4 Changes to the A2Geant4 Program

To complete this project, I wrote three additional classes for the simulation and made edits to many others. These changes are presented in Tables 4 and 5. A few additional files, containing information or analysis scripts, I have also added to the folder, as shown in Table 6.

Name	Base class	Description
A2TPC	A2Target	ACTAM TPC as simulated target
A2ElectricField	G4ElectricField	Electric field for use in TPC
A2DriftModel	G4VFastSimulationModel	Custom definition of electron drift

Table 4: Classes added to the **A2Geant4** program to describe the TPC.

Class	Changes made
A2DetectorSetup	Added option to use A2TPC as a target
A2Hit	Added option to record charge in a hit
A2SD	Added option to record charge in a sensitive detector, customized how event time is recorded for the TPC anode
A2CBOutput	Added branches for TPC data as well as true kinetic energy of particles in branch *klab*
A2SteppingAction	Added code to call A2DriftModel when propagating a low energy electron through the active volume of the TPC
A2	Registered G4EmLivermore (low energy electromagnetic physics) and G4FastSimulationPhysics (on which the electron drift model is built) with the overall physics list

Table 5: Changes made to existing classes of **A2Geant4** to incorporate the ACTAM TPC.

Name	Directory	Purpose
ACTAM-01-22.pdf	docs	Documentation of ACTAM TPC
TPC.dat	data	TPC dimensions used in runtime
TPCcalibrate.mac	macros	Geant4 macro that generates files to be used in charge-energy calibration
Calibrate.C	out	CERN ROOT script to calibrate charge-energy relationship
Reconstruct.C	out	CERN ROOT script to reconstruct event parameters from TPC output
change_log.txt	.	Chronological log of all changes made

Table 6: Additional files present in my branch of **A2Geant4**.

5 Studies in GoAT

I have implemented the TPC readout in both **acqu** and **a2GoAT**, but the current **a2GoAT** analysis is very preliminary. This is done in the class *Analyze*. All branches of data produced by the simulation are read into the class, and analysis of this starts at line 558 of *src/PAnalyze.cc*. The names of variables change slightly between the two programs, but the data processing in **a2GoAT** is much the same as what is produced by the standalone CERN ROOT script *Reconstruct.C*.

Name	Data
TPC_Z	Reconstructed z coordinate of event position
TPC_Energy	Reconstructed kinetic energy of recoil particle
TPC_Theta	Reconstructed polar angle of recoil particle

Table 7: Histograms produced by a2GoAT using the ACTAM TPC output.

I recommend future work to focus on the part of this project. While the electron production rate is incorrect, useful results can still be obtained. For feasibility studies, the ACTAM TPC simulation can be used to process event generator files with different run-time parameters. A useful next step would be to study missing energies and angles in **a2GoAT** with the reconstructed data. Updates done to the event reconstruction framework would also be most useful if implemented in the *Analyze* class of **a2GoAT**.

6 To Run

Download my **A2Geant4**, **acqu**, and **a2GoAT** from GitHub (for **acqu** and **a2GoAT**, there is a separate branch named *tpc*). In **A2Geant4**, ensure the target is set to *TPC* inside *detectorSetup.mac*, then run as normal. In **acqu**, run with *data.TPC/MC-CBTaggTAPS.dat* to pass along the simulated data. In **a2GoAT**, use the *Analyze* class with *configfiles/Analyze-TPC.dat.*

When changing active gas type, target pressure, or anode design, make sure that the same variables are set in *A2Geant4/data/TPC.dat* and *a2GoAT/configfiles/Analyze-TPC.dat.*

References

[1] F. Metzger. *TPC-signal simulation.* Presentation slides. 2020.

[2] Geant4 Collaboration. *Book for Application Developers, Release 10.3.* 2017.

[3] P. Martel. Personal correspondence (many emails where I asked for help with software). 2021.